

PortfolioCheck — Automated Verification of Formal Coursework Submission Requirements

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Solution and Concept

Every portfolio submission is governed by formal provisions that are entirely mechanical: a naming pattern for the main folder, four subdirectories with exact names, permitted file formats, a size limit of 15 MB, a links file, and a code archive that must be identical to the public GitHub repository. These provisions carry marks, yet students verify them by hand in the last hours before a deadline, and errors surface only when they can no longer be corrected. Research on automated assessment concentrates on the correctness of program code; the formal envelope of a submission has received almost no attention.

PortfolioCheck closes this gap. A student uploads the zip folder they intend to submit and selects the portfolio phase. The web application evaluates the archive against seven rules derived from the assignment brief and returns a report in which every rule is marked *passed*, *to review* or *must fix*. Two principles define the concept. First, every negative verdict states the exact correction required — a report that only says “wrong” would reproduce the student’s problem. Second, every finding names the provision of the brief it derives from, so the student can check the verdict against the source. The tool judges form, never content, and stores nothing: an upload is deleted before the response is returned. The primary target group is students in distance-learning programmes, above all those submitting for the first time and those working close to a deadline; a student starting from nothing can also download an empty, correctly named folder structure.

Making of: Technical Approach

The project followed an incremental Kanban approach in GitHub Projects, with a work-in-progress limit of one and weekly reviews; Scrum was rejected because its events presuppose a team. Eleven work packages across the three portfolio phases were each linked to the requirements they deliver: thirteen functional and nine non-functional requirements, the latter structured along ISO/IEC 25010.

The architecture is layered around a domain model that depends on nothing. Each formal provision is one rule class registered with a rule registry, so a new rule changes no other component — a property demonstrated when the final rule was added. The implementation uses Python 3.11 and FastAPI with server-rendered templates; Django was set aside because its database and administration features would remain unused, and a single-page front end because one form and one report do not justify a second build pipeline. Uploads are treated as hostile: before anything is extracted, archives with path traversal, absolute paths, excessive entry counts or extreme compression ratios are rejected. The application runs as an unprivileged Docker container on a read-only filesystem and is deployed on Render.

The most distinctive rule verifies that the public repository and the submitted code archive are identical — a provision practically impossible to check by hand. It reads the repository address from the links file, retrieves the file tree through the GitHub REST API, and compares both sides using git blob hashes, the content fingerprints git itself uses. Missing, surplus and modified files are

therefore all detected without downloading any file, while Windows line endings and the wrapper folder of GitHub's "Download ZIP" are correctly ignored. If GitHub cannot be reached, this rule degrades to a warning and all other rules still run.

Version and release control were kept in a single public repository: the source code, the test suite, the sample data and the architecture documentation are versioned together, and each portfolio milestone is marked by a release tag, so documentation and implementation cannot drift apart.

Testing and Results

The application comprises about 1,100 lines of code and 900 lines of tests. All 73 automated tests pass; they cover every rule against a compliant and a violating submission, the extraction defences, the GitHub client against a simulated service, and the complete path through the web interface. No test uses the network. Thirteen manual test cases specify concrete input files and the exact expected output; nine realistic sample submissions, generated reproducibly by a script, serve as their input and double as the test data that makes the application usable immediately. All thirteen functional requirements are implemented, and every non-functional requirement has recorded evidence: a typical submission is checked well within the three-second target, a deliberately failing rule leaves the rest of the report intact, and a single container image, configured only through environment variables, serves both local use and the cloud deployment. The documented architecture matches the code: every component maps to a named module, and every dependency in the UML diagram corresponds to an import. Seven items of technical debt remain documented, among them a global rule registry and hard-coded limits.

Lessons Learned

Three lessons stand out. First, *defences before features*: implementing the archive safety checks before any rule meant no untrusted input ever passed an unhardened path, and nothing had to be retrofitted. Second, *isolate the external dependency early*: placing GitHub behind an interface let the rest of the system proceed independently and made the most complex rule fully testable offline. Third, *the feedback loop is part of the method*: the tutor's feedback corrected the architecture notation to UML, clarified that technical debt means implementation shortcuts rather than missing features, and showed that a Kanban board is evidence only when it lives in a tool linked to the commits, not when it is drawn afterwards. Writing each test together with its rule proved to be the single most effective practice: when the final rule was added late in the project, the existing suite confirmed immediately that nothing else had broken.

Regarding time and resources, the repository comparison consumed the largest share of the final phase, as the risk register had anticipated. Scheduling it first in that phase left enough buffer for the deployment, and keeping every increment independently deliverable meant the project never depended on unfinished work.

Outlook

The most valuable next steps follow from the recorded technical debt. An explicit rule registry and externalised configuration would allow the rule set to be adapted to the briefs of other courses and institutions, and a cache for repository inventories would let one shared deployment serve a whole cohort in the days before a deadline. A further rule could check the abstract's page count, font and margins.